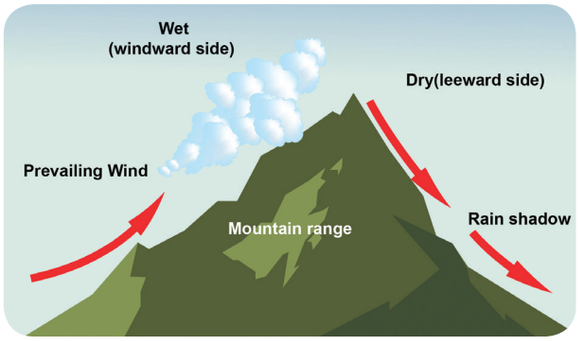


Cloud Cover

Clouds reflect a large amount of radiation from the sun. This prevents the entry of heat to the earth's surface. So, in areas generally of cloudless sky like the deserts, temperature is very high. On the other hand under cloudy sky, the temperature is low.

Ocean currents

The warm ocean currents raise the temperature of the nearby coastal areas, while the cold current lower the temperature of a place.

Natural vegetation

The trees release water vapour into the air and makes it cool. Thus forest areas have lower range of temperature throughout the year in contrast to non-forested areas.

3.4 Winds

The horizontal movement of air along the surface of the earth is called the "**Wind**" while the vertical movement of air is a called an "**Air Current**". The winds always blow from a high pressure area to a low pressure area. Wind is mostly named after the direction from which it blows. For example, the wind blowing from the east is known as the easterly wind or easterlies.

An "**anemometer**" records **wind speed** while a "**wind vane**" measures the **direction of the wind**. The unit of measurement is **kilometre per hour or knots**



Anemometer



Wind vane

Types of Winds

Winds are generally classified into the following four major types:

- Planetary winds
- Periodic winds
- Variable wind
- Local wind

Planetary winds:

The winds which constantly blow in the same direction throughout the year are called the Planetary winds. They are also called as permanent winds or the prevailing winds. These winds include **Trade winds, Westerlies** and **Polar Easterlies**

Trade Winds

Trade winds blow from the subtropical high pressure belt to the Equatorial low pressure belt in both the hemispheres. They blow with great regularity, force and in a constant direction throughout the year. These winds were very helpful to traders who depended on the winds while sailing in the seas. And so, they are named as Trade winds.

Activity

Find the correlation between the Trade Winds and the location of prominent deserts like Sahara, Atacama etc.

Westerlies

Westerlies are the permanent winds that blow from the tropical high pressure belt to the sub polar low pressure belt in both the hemispheres. They blow from South West to North East in the northern hemisphere and North West to South East in the southern

9th_Geography_Unit_3_EM.indd 184 4/4/2022 5:28:44 PM